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Lessons

Using Substitution

A **system of equations** is a collection of equations that use the same variables. In a system, a variable represents the same quantity in each equation.

Suppose the area of a rectangle is 40 square inches and the perimeter of the same rectangle is 28 inches. Area and perimeter are found using length (L) and width (W).

You can use this information to write a system about the rectangle. Both equations use the same variables, and both are about the same rectangle. The brace shows the equations are part of a system.

brace → $\begin{cases} L \bullet W = 40 & ① \\ 2L + 2W = 28 & ② \end{cases}$

Area = Length × Width

Perimeter = 2 × Length + 2 × Width

$\begin{cases} 10 \bullet 4 = 40 \\ 2(10) + 2(4) = 28 \end{cases}$

$LW = 40$

$W = \frac{40}{L}$ → substitute into ② for W.

$2L + 2\left(\frac{40}{L}\right) = 28$

$L\left(2L\right) + L\left(\frac{80}{L}\right) = (28)L$

$2L^2 + 80 = 28L$

$2L^2 - 28L + 80 = 0$

then isolate L

(we'll return this idea later).

The values for L and W that make the first equation true will also make the second equation true. This system is true when $L = 10$ and $W = 4$.

Solving a system means finding the values of the variables. One way to solve a system is by graphing the equations.

Example 1

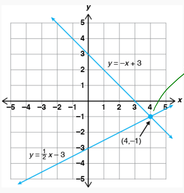
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Solve: $\begin{cases} y = -x + 3 \\ y = \frac{1}{2}x - 3 \end{cases}$ Linear equations of the form $y = mx + b$ → y-intercept, slope

Step 1
Graph both equations on the same grid.
Remember, you can always graph a line by making a table of values and then plotting points.

$y = -x + 3$	
x	y
−1	4
0	3
1	2

$y = \frac{1}{2}x - 3$	
x	y
−2	−4
0	−3
2	−2



Step 2
The lines intersect at point (4,−1). The x- and y-coordinates of this point are the solution to the system: $x = 4$ and $y = -1$.

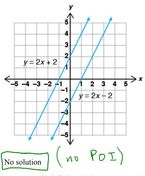
Step 3
Check your answer. Substitute the values into the system. Make sure both equations are true.

$y = -x + 3$
$-1 = -(4) + 3$
$-1 = -1$ True

$y = \frac{1}{2}x - 3$
$-1 = \frac{1}{2}(4) - 3$
$-1 = 2 - 3$
$-1 = -1$ True

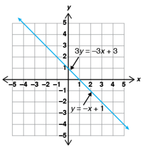
Most systems have one solution that is written as an ordered pair. In special situations, a system may have no solution or an infinite number of solutions.

On a graph, a system with no solution has parallel lines that never intersect.



No solution (no POI)

A system with infinite solutions appears to be one line because both lines occupy the same space. Every point on one line coincides with a point on the other line.



Infinite solutions

The **substitution method** for solving a system substitutes an expression from one equation into the other.

Example 2

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Solve: $\begin{cases} 3x - y = -2 & ① \\ y = 18 - 5x & ② \end{cases}$

Look at the second equation: $y = 18 - 5x$. It states that the variable y is equal to the expression $18 - 5x$. You can substitute this expression for y in the first equation.

Step 1
Substitute $18 - 5x$ for y in the first equation. → ①

$3x - y = -2$
$3x - (18 - 5x) = -2$

Step 2
Solve for x.

$3x - (18 - 5x) = -2$
$3x - 18 + 5x = -2$
$8x - 18 = -2$
$8x = 16$
$x = 2$

Step 3
Substitute the value for x in the second equation and solve for y. → ②

$y = 18 - 5x$
$y = 18 - 5(2)$
$y = 18 - 10$
$y = 8$

Write the solution as an ordered pair. The solution is **(2, 8)**

Check: Make sure the solution works in both equations in the system.

$3x - y = -2$
$3(2) - (8) = -2$
$6 - 8 = -2$
$-2 = -2$ True

$y = 18 - 5x$
$8 = 18 - 5(2)$
$8 = 18 - 10$
$8 = 8$ True

The solution is correct since it makes both equations true.

Sometimes you will need to rewrite one of the equations to isolate a variable. You can isolate either x or y.

Example 3

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Solve: $\begin{cases} x - 2y = -2 \\ 8x - 7y = -16 \end{cases}$

Look for a variable that would be easy to isolate. The first equation has an x without a number next to it. Isolate the x.

$x - 2y = -2$
$x = 2y - 2$

Step 1
Substitute $2y - 2$ for x in the second equation.

$8x - 7y = -16$
$8(2y - 2) - 7y = -16$

Step 2
Solve for y.

$8(2y - 2) - 7y = -16$
$16y - 16 - 7y = -16$
$9y - 16 = -16$
$9y = 0$
$y = 0$

Step 3
Substitute 0 for y in the first equation and solve for x.

$x - 2y = -2$
$x - 2(0) = -2$
$x - 0 = -2$
$x = -2$

The solution is **(−2, 0)**.

Key Terms

elimination method
substitution method
system of equations

Tip

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It's very important that you use both equations when you are solving a system. If you isolate the variable in the *first* equation, substitute the expression into the *second*. Then put the solution from that step back into the *first* equation. The equations are related, so you need to involve both to find the answer.

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Lesson 4: Systems of Equations

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